



Azure Arc: Optimizing Server Management

Free or "no additional cost" services
enabled by Azure Arc for enterprise
customers

John Kelbley - Director
Azure Advanced Migration
johnkel@microsoft.com



What is Azure Arc?

Arc is Secure Management Dial Tone



Reach

Windows and Linux
VM and bare metal,
At scale searchable inventory



Configure

Consistent VM extensions
Centralized agent management—
Monitoring, Security, Update Management



Govern

Built-in Azure policies
Compliance across environments
Audit and enforce OS settings



Secure

Azure Active Directory
Managed Identity
Server security baselines



One consistent Azure platform for hybrid and multicloud

Consistent Operations,
Governance, and Security



Azure Arc-enabled
application services



Azure Arc-enabled
data services and ML



Azure Arc-enabled infrastructure



Hybrid and Multicloud



Arc Capabilities by Cost / Entitlement

No Cost Capabilities
("free")

For Fee...or "Included"
with Software Assurance

Only Available
with Software Assurance

For Fee "Standalone"

Customers With
Software Assurance on
Windows and SQL can
use most Arc enabled
capabilities at
NO ADDITIONAL COST!

“No Cost” Arc Enabled Capabilities

Azure Arc-Enabled
Servers: “Free” Stuff! –
Azure Advanced Migration



SSH / RDP Access

Run Command

System Managed Identity

Tagging and inventory management

Microsoft Entra ID Authentication

Role Based Access Control (RBAC)

Azure Resource Manager

Windows Admin Center

Azure Migrate Assessments & Business Case

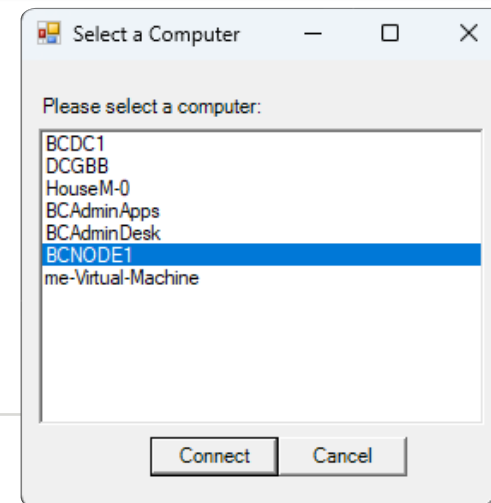
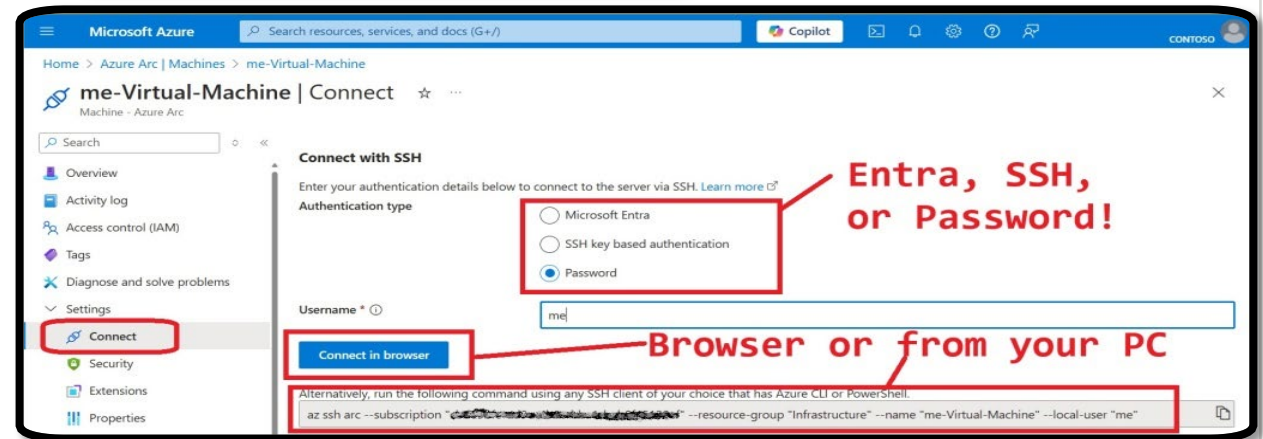
Site Manager (preview)

No Cost Highlight: Secure SSH / RDP Access

Azure Benefits

- Arc connected system can be securely accessed remotely
- Secure Linux can use Entra ID with out the need for local
- Additional automation can be to simplify access or allow “just in time” enablement of SSH / RDP services – checkout [Easy\(er\) Server Remote Access with Arc – Azure Advanced Migration](#)

Customer Experience: Secure SSH / RDP



No Cost Highlight: Remote Command Execution

Azure Benefits

- Securely execute commands / scripts on remote Arc connected system
- Use Entra ID with out the need for local administrator accounts
- Can leverage API, Azure CLI, or PowerShell [Arc Remote Run Command – Azure Advanced Migration](#)

Customer Experience: Remote Command Execution

Examples for remotely executing “Shutdown” via PowerShell and the Azure CLI

```
Connect-AzAccount
New-AzConnectedMachineRunCommand -ResourceGroupName
$AZResourceGroup -SourceScript "Stop-Computer -ComputerName
localhost -Force" -RunCommandName "Shutdown" -MachineName
$TargetMachine -Location "eastus" -NoWait
```

```
#####
# Be sure you have the extensions loaded and updated
#####
# az upgrade --allow-preview True
# az config set auto-upgrade.prompt=no
# az extension add --name connectedmachine

az login
az connectedmachine run-command create --name "Shutdown" --
machine-name $TargetMachine --resource-group $AZResourceGroup --
script "Stop-Computer -ComputerName localhost -Force" --no-wait
```

Windows Admin Center in Azure for Arc

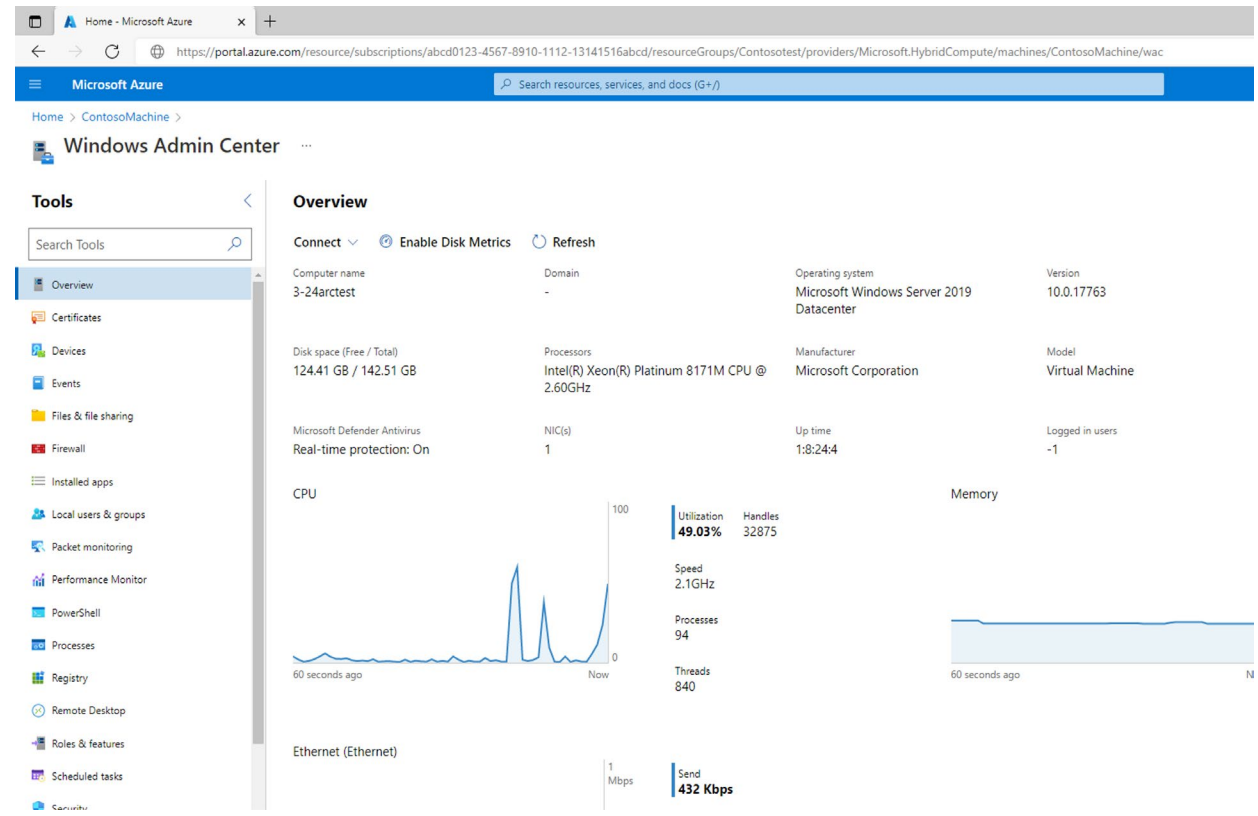
Capabilities

Securely manage your servers from anywhere without a VPN, public IP address, or other inbound connectivity

Management, configuration, troubleshooting, and maintenance functionality in a **single pane of glass**

- Virtual machines tool
- Event viewer
- Remote Desktop
- File explorer
-and much more!

Customer Experience



For Fee... or “Included” with Software Assurance

[Windows Server
Management enabled by
Azure Arc - Azure Arc |](#)

[Microsoft Learn
Overview - SQL Server
enabled by Azure Arc |
Microsoft Learn](#)

Azure Change Tracking and Inventory
\$6 / VM / Month with Azure Policy
or NO COST with Windows Server SA

Azure Policy Machine Guest Configuration
\$6 / VM / Month with Azure Change Tracking
or NO COST with Windows Server SA

Azure Update Manager
\$5 / VM / Month...
or NO COST with Windows Server SA

Azure Change Tracking and Inventory

Azure Benefits

- Discover what's installed on your machines and their changes
 - Software
 - Services/Daemons
 - Files
 - Registries
- View how many machines have a specific software or version
- Updated when a change is detected
- View changes when troubleshooting
- Alerting on critical changes
- Differences in inventory snapshots

Requires AMA (Azure Monitor Agent) with Log Analytics workspace

Customer Experience: Activation

Microsoft Azure (Preview)

arc-dp-wus | Inventory

Machine - Azure Arc | Directory: Microsoft

Search

Open change and inventory center | Refresh | Log Analytics | Settings | Feedback

Operations

- Policies
- Machine Configuration
- Run command (preview)
- SQL Server Configuration
- Updates
- **Inventory**
- Change tracking

Licenses

Windows management

- Remote Support (preview)
- Windows Admin Center (preview)
- Disaster recovery (preview)
- Best Practices Assessment (preview)

Monitoring

- Insights
- Logs
- Workbooks

Automation

- CLI / PS
- Tasks

Changes tracking with AMA

Tracks changes in virtual machines hosted in Azure, on-premises, and other cloud environments to help you pinpoint operational and environmental issues. [Learn more](#)

Software(0) | Files(0) | Windows Registry(0) | **Windows Services(205)**

Search to filter items...

Service	Display Name	Status	Start up type	Last Refreshed Time
AdtAgent	Microsoft Monitoring Agent Audit Forwarding	Stopped	Disabled	10/30/2024, 9:46 AM
AJRouter	AllJoyn Router Service	Stopped	Manual	10/30/2024, 9:46 AM
ALG	Application Layer Gateway Service	Stopped	Manual	10/30/2024, 9:46 AM
AppIDSvc	Application Identity	Running	Auto	10/30/2024, 9:46 AM
AppInfo	Application Information	Stopped	Manual	10/30/2024, 9:46 AM
AppMgmt	Application Management	Stopped	Manual	10/30/2024, 9:46 AM
AppReadiness	App Readiness	Stopped	Manual	10/30/2024, 9:46 AM
AppVClient	Microsoft App-V Client	Stopped	Disabled	10/30/2024, 9:46 AM
AppXSvc	AppX Deployment Service (AppXSVC)	Stopped	Manual	10/30/2024, 9:46 AM
AudioEndpointBuilder	Windows Audio Endpoint Builder	Stopped	Manual	10/30/2024, 9:46 AM
Audiosrv	Windows Audio	Stopped	Manual	10/30/2024, 9:46 AM
AxInstSV	ActiveX Installer (AxInstSV)	Stopped	Disabled	10/30/2024, 9:46 AM
BDESVC	BitLocker Drive Encryption Service	Stopped	Manual	10/30/2024, 9:46 AM
BFE	Base Filtering Engine	Running	Auto	10/30/2024, 9:46 AM
BITS	Background Intelligent Transfer Service	Stopped	Manual	10/30/2024, 9:46 AM

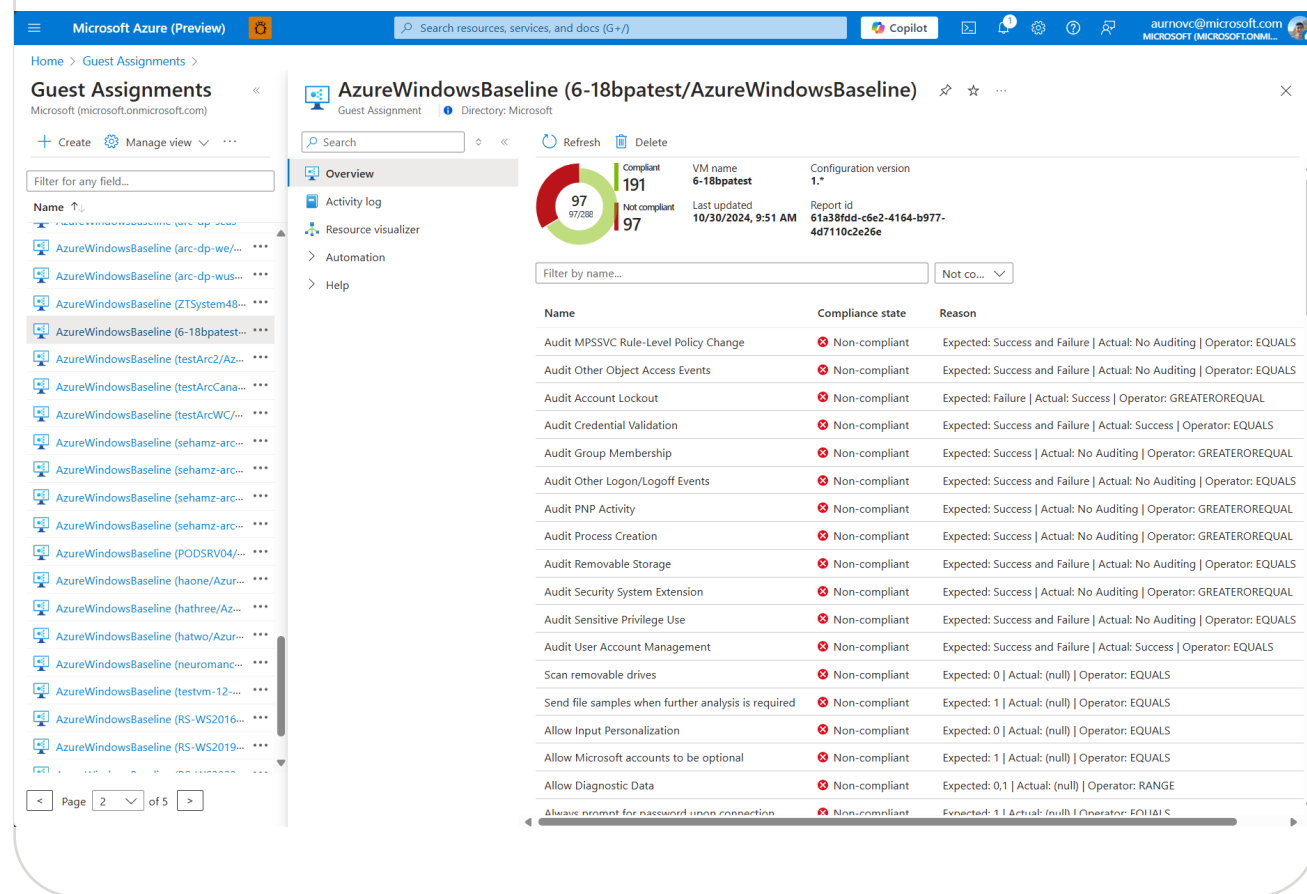
Showing 1 - 15 of 205 results.

Azure Machine Configuration

Azure Benefits

- Configuration as code defining machine properties:
 - Operating System settings
 - App configuration or presence
 - Environment settings
- Advanced reporting through Azure Resource Graph
- Ability to create custom packages
- Built in configurations for operational and security best practices
 - OS settings, expiring certificates, networking capabilities
 - TLS Versions, local user accounts, security baselines
- Native to Azure Arc's Connected Machine Agent

Customer Experience: Activation

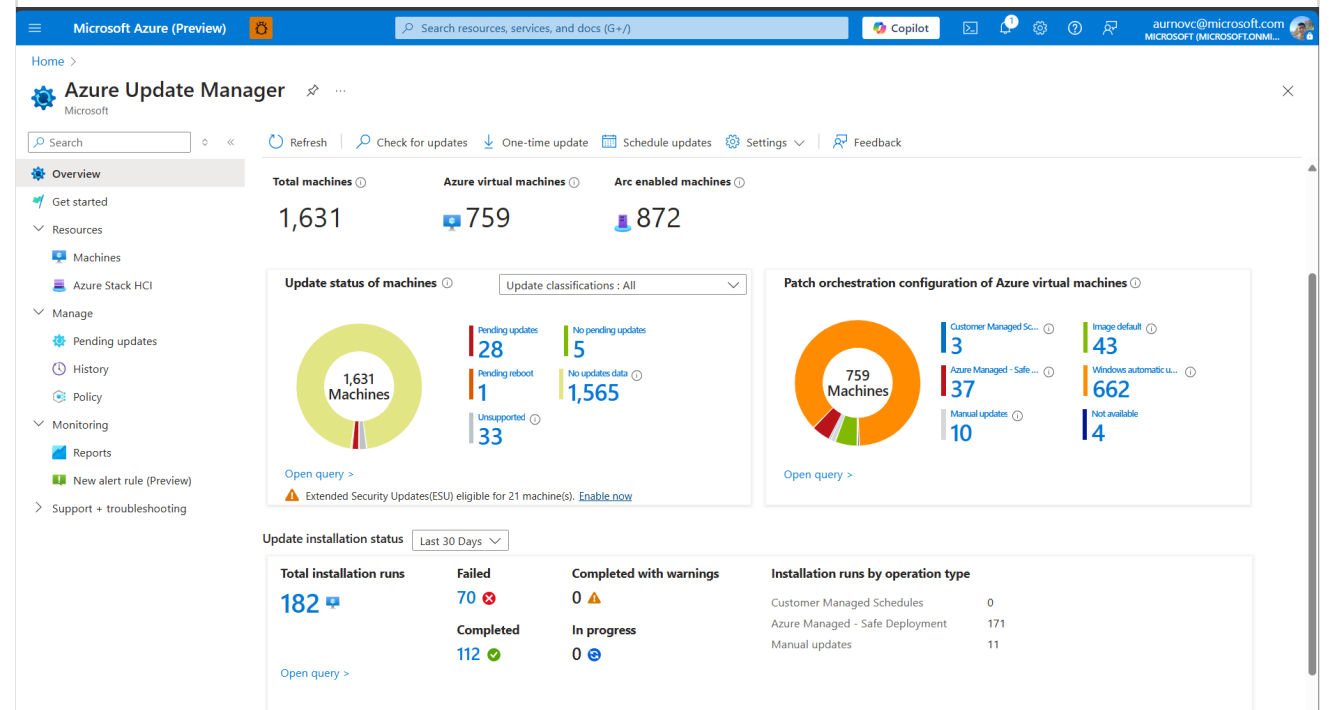


Azure Update Manager

Capabilities

- Assess the update status of servers across your Azure, on-premises, and multi-cloud estates from one place
- Easily deploy updates to your machines
 - Schedule one-off or recurring updates
 - Schedule recurring updates for machine groups
 - Target systems by name, imported groups, or custom query
- Verify update compliance across your environment
- Track update deployments across your environment
- Audit past deployments

Customer Experience



Only Available with Software Assurance

Windows Server
Management enabled by
Azure Arc - Azure Arc |

Microsoft Learn
Overview - SQL Server
enabled by Azure Arc |
Microsoft Learn

SQL Best Practice Assessment

SQL Single Sign on with Entra ID

SQL Auto Patching

SQL Auto-Backup

SQL TDE with Azure Key Vault

Windows Server Best Practices Assessment

Windows Server Remote Support

Windows Server Disaster Recovery*

Windows Server Network HUD/Accelnet (WS 2025)

*Additional cost for storage and compute failover

WS Management enabled by Azure Arc

Azure Benefits

Customers with WS Software Assurance or enrolled in Windows Server Pay as you Go can access the following capabilities at no additional cost:

- Azure Update Manager
- Azure Machine Configuration [Policy]
- Azure Change Tracking and Inventory*
- Best Practices Assessment*
- Remote Support
- Network HUD
- Azure Site Recovery configuration*
- Windows Admin Center in Azure for Arc

*May incur additional storage, log data ingestion, and compute costs

Customer Experience: Activation

The screenshot shows the Microsoft Azure portal interface. The main navigation pane on the left includes 'Overview', 'All Azure Arc resources', 'Site manager', 'Azure Arc resources', 'Host environments', 'Data services', 'IoT', 'Application services', 'Licenses', 'Extended Security Updates', 'Windows Server', 'Management', and 'DevOps'. The 'Windows Server' section is selected. The main content area displays 'Azure Arc | Windows Server licenses' with tabs for 'Azure benefits' and 'Pay-as-you-go'. Below the tabs, there's a table of licenses:

Name	Benefits	Arc agent status	Resource
arc-server-01	Not activated	Connected	contoso
arc-server-02	Not activated	Connected	contoso
arc-server-03	Not activated	Connected	contoso

On the right, the 'Activate Azure benefits' dialog is open. It contains the following text:

Activate Azure benefits

If your machine(s) have Windows Server licenses with active Software Assurance or such licenses are active subscription licenses, you can access key management services at no additional cost. [Learn more](#)

Benefits

You can now set up any of the following services

- Azure Update Manager
- Change Tracking and Inventory
- Azure Machine Configuration
- Windows Admin Center
- Remote support
- Advanced networking (WS 2025 only)
- Best practices assessment*
- Disaster recovery*

*Any Azure compute or storage usage will still incur costs

☒ By checking this box, you attest that your Windows Server licenses have active Software Assurance or your Windows Server licenses are active subscription licenses.

Activate **Cancel** [Give feedback](#)

Windows SA: Remote Support

Capabilities

Get **professional support** by permitting access to your machine remotely

Grant consent while controlling access level and duration of access (JIT) on an incident-by-incident basis

View detailed transcripts of all executed operations at any time

Revoke consent at anytime

Access is auto disabled once the consent duration expires

Customer Experience

The screenshot displays the Microsoft Azure portal interface for a virtual machine named 'prgokhal-vm1'. The left sidebar shows the navigation menu with 'Remote Support (preview)' selected. The main content area is titled 'prgokhal-vm1 | Remote Support (preview)' and includes a search bar and action buttons: 'Revoke access', 'Configure access', and 'Refresh'. Below these, the 'Access state' is shown as 'Granted' with a 'Microsoft support access...' link. A section titled 'Remote Support Sessions' shows a table of past sessions. To the right, a 'Configure remote support access' dialog is open, showing options for 'Access level' (Diagnostics or Diagnostics and repair) and 'Access expires on' (Mon Nov 04 2024 12:14 PM). The dialog has 'Save changes' and 'Cancel' buttons.

Session start time	Session end time	Server
10/3/2024, 11:11:11 AM	10/3/2024, 1:11:14 PM	PRGOKHAL-VM1
10/2/2024, 12:17:53 PM	10/2/2024, 12:23:34 PM	PRGOKHAL-VM1
10/1/2024, 10:28:59 AM	10/1/2024, 10:29:29 AM	PRGOKHAL-VM1

Windows SA: Best Practices Assessment

Capabilities

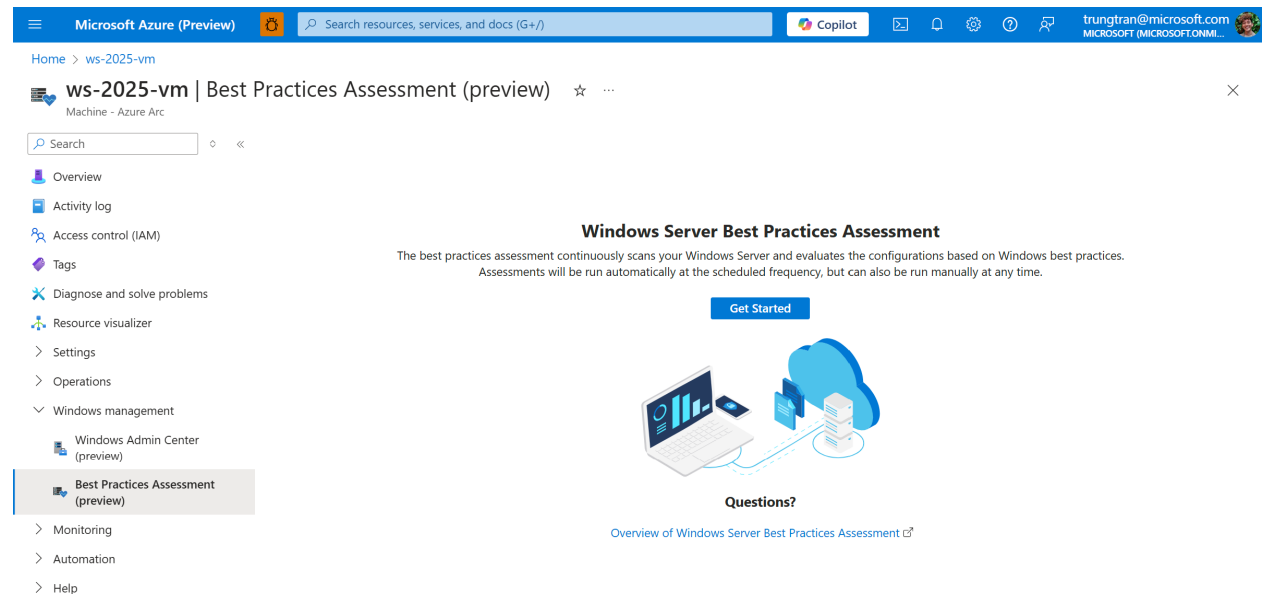
Collects and analyze server data to generate list of issues and **remediation guidance**

Provide customers with best practices to **improve performance** of their server infrastructure and features such as deploying applications, software updates, etc...

Assesses in the following areas:

- Server Baselines
- Server Security
- Hyper-V
- Failover Cluster
- IIS

Customer Experience



Windows SA: Azure Site Recovery configuration

Capabilities

Installs the Azure Site Recovery agent on your machine and associates the replication policy, vault, and Hyper-V site with the agent

Ensure business continuity

Keeps business apps and workloads **running during outages**

Provides **replication** and data **resilience**

Customer Experience

The screenshot displays the 'Create Recovery Services vault' wizard in the Microsoft Azure portal. The interface is split into two main sections: 'Prepare infrastructure' on the left and 'Project Details' on the right.

Prepare infrastructure:

- Vault to prepare infrastructure for ***: A dropdown menu with the placeholder text 'selected vault name will be shown here...'. Below it are links for 'Select existing vault' and 'Create new vault'.
- Hyper-V site**: A dropdown menu with the placeholder text 'selected site name will be shown here...'. Below it are links for 'Select existing hyper-v site' and 'Create new site'.
- Replication policy**: A dropdown menu with the placeholder text 'selected policy name will be shown here...'. Below it are links for 'Select existing policy' and 'Create new policy'.

Project Details:

Select the subscription and the resource group in which you want to create the vault.

- Subscription ***: A dropdown menu showing 'SME Demos, Testing - Microsoft Azure Internal Consumption'.
- Resource group ***: A dropdown menu showing 'theo-rg' with a 'Create new' link below it.

Instance Details:

- Vault name ***: A text input field with the placeholder text 'Enter the name for your vault.'
- Region ***: A dropdown menu showing 'East US'.

A blue information banner at the bottom of the right pane states: 'Cross Subscription Restore is enabled by default for all vaults. Visit vault 'Properties' to disable the same. [Learn more.](#)'

At the bottom of the wizard, there are two buttons: 'Cancel' and 'Prepare infrastructure' (disabled) on the left, and 'Review + create' (active) and 'Next: Redundancy' (disabled) on the right.

[Tutorial to set up Azure VM disaster recovery with ...](#)

Windows SA: Network HUD

Capabilities

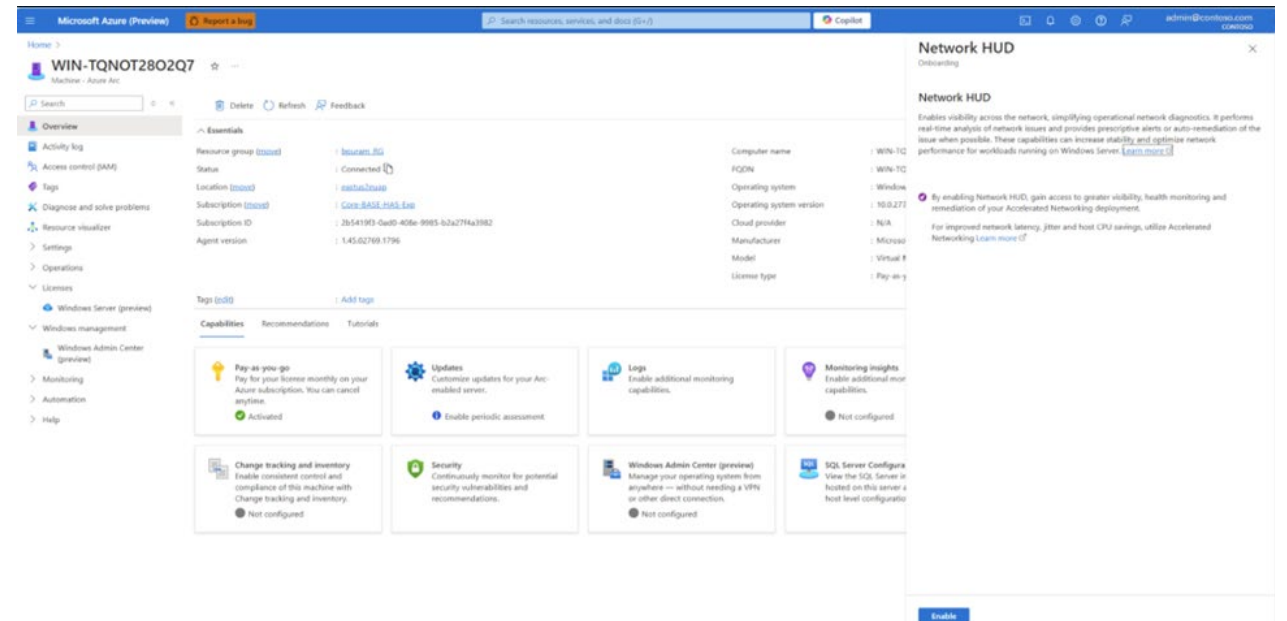
Identify misconfiguration on the physical network.

Identify operational issues with physical hardware (NICs, cabling, physical switch).

Identify and optimize your configuration for increased performance.

Auto-remediate some host issues to increase stability and performance.

Customer Experience



Arc Enabled Services: For Free “Stand alone” Cost



Defender for Cloud / Defender for SQL
\$0.015/vCore/hour

Azure Monitor & Log Analytics
\$2.30/GB [Pay as you go]

SQL Purview Integration

SQL Synapse Link

Extended Security Updates
(Windows Server 2012, SQL Server 2012)

Sentinel

First Step: Installing the Arc Agent

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the Microsoft Azure logo, a search bar, and various icons. The main content area is titled 'Add servers with Azure Arc' and includes a breadcrumb trail: 'Home > Azure Arc | Machines >'. Below the title, there's a sub-header 'Machines - Azure Arc' and a paragraph explaining that Azure Arc allows managing on-premises servers and servers from other clouds. A 'Learn more' link is provided. The page is divided into five cards, each representing a different method to add servers. Each card has a title, a description, and a button to proceed.

Home > Azure Arc | Machines >

Add servers with Azure Arc

Machines - Azure Arc

Azure Arc allows you to use Azure tools to manage on-premises servers and servers from other clouds. We'll start with some prerequisites and deploy the Azure Connected Machine agent. [Learn more](#)

Add a single server

This option will generate a script to run on your target server. The script will prompt you for your Azure login, so this option is best for adding servers one at a time.

[Generate script](#) [Learn more](#)

Add multiple servers

To add multiple servers to Azure, we will generate a script that handles authentication through a service principal. You will see that and other prerequisites next.

[Generate script](#) [Learn more](#)

Add Windows Server with installer

Onboard a single Windows Server with your Azure credentials using an installer that guides you through the process step by step.

[Download installer](#) [Learn more](#)

Add servers from AWS

Connect your Amazon Web Services (AWS) accounts for scalable and automatic onboarding of EC2 instances by creating an AWS connector.

[Add servers](#) [Learn more](#)

Add servers from Update Management

Non-Azure servers managed by the Update Management service can be easily connected to Azure via Azure Arc. Once you have selected the servers, the deployment will happen automatically.

[Add servers](#) [Learn more](#)

dd

Pro Tip: Use Arc to Install Arc Agent!



[Deploying Azure Arc with Azure Arc – Azure Advanced Migration \(wordpress.com\)](#)

- Prerequisites:
 - At least one Windows Server already onboarded to Azure Arc which resides in the environment where the servers are located that you wish to onboard
 - An Azure Key Vault to securely manage and store the required 'secrets' like the local administrative account and password to use on the target server(s) as well as the SPN ID and SPN secret that can onboard servers to the desired Azure Resource Group/Subscription/Tenant.
 - An Entra credential which has rights to the already Azure Arc-enabled Windows Server to allow the execution of custom script extensions, run command, or both.
 - The system-assigned identity of the Azure Arc-enabled machine with rights to retrieve secrets from the Azure Key Vault that holds them.
 - A script to orchestrate it all: [kevinsul/arcserver-onboard-w-managed-id \(github.com\)](#)

Pro Tip: Use PowerShell to check the box!



Part of the process is to attest whether the server has Software Assurance. A tedious task of checking the SA box.

This can be automated using PowerShell now.

[Attestation of SA using PowerShell Scripting](#)

Thank You!

Learn more about Azure Virtual Desktop – Additional sessions Today!

- 10:30AM – 11:00AM
- 12:45PM – 1:30PM
- 1:45PM – 2:30PM
- 3:00PM – 3:45PM

Introduction to Azure Virtual Desktop
Azure Virtual Desktop: Real World Use Cases
Deploying and Managing AVD
Azure Arc: Optimizing Server Management

- 5:30PM – 6:30PM

Happy Hour sponsored by  **nerdio**

John Kelbley - Director
Azure Advanced Migration
johnkel@microsoft.com

Shawn Meyer – Principal CSA
Microsoft Federal
shawn.meyer@microsoft.com



FEDERAL AZURE SUMMIT AGENDA

JAN 16
2025

8:30 - 9:00

Registration and Welcome Coffee

9:00 - 9:15

Opening Remarks by [Javier Vasquez](#)

9:15 - 9:45

Keynote: Future of Hybrid Cloud by [Jason Payne](#)

9:45 - 10:15

Adaptive Cloud: Extending Cloud Capabilities to the Edge by [Eric Sanchez](#) / [Joe Fox](#)

10:15 - 10:30

Morning Break

10:30 - 11:00

Introduction to Azure Virtual Desktop (AVD) by [John Kelbley](#) / [Shawn Meyer](#)

11:00 - 11:45

Introduction to Azure VMWare Solution by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

11:45 - 12:45

Lunch Sponsored by



12:45 - 1:30

[Session 1A: Real-World Use Cases of Azure Virtual Desktop](#) by [John Kelbley](#) / [Shawn Meyer](#)

[Session 1B: Real-World Use Cases of Azure Edge Devices](#) by [Rich Shillington](#) / [Jason Benson](#)

1:45 - 2:30

[Session 2A: Deploying and Managing AVD](#) by [Shawn Meyer](#)

[Session 2B: Real-World Use Cases of Azure VMWare Solutions](#)
by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

2:30 - 2:45

Afternoon Break

3:00 - 3:45

[Session 3A: Azure Arc – Optimizing Server Management](#) by [John Kelbley](#)

[Session 3B: Deploying & Managing Azure VMWare Solutions](#)
by [Ruth Maygar](#) / [Kevin Jellow](#) / [Connor Loving](#)

4:00 - 4:45

Fed Chief Architect Panel: The Future of Cloud and Edge Computing by [Jason Payne](#) / [Chief Architects](#)

4:45 - 5:00

Closing Remarks and Networking by [Javier Vasquez](#)

5:30 - 6:30

Happy Hour Sponsored by



Please visit <https://aka.ms/FedAzSummit/home> for more information